Article

International Relations
2019, Vol. 33(2) 286–303

DOI: 10.1177/0047117819834629
journals.sagepub.com/home/ire
SAGE

Technological change and international relations

Daniel W Drezner
Tufts University

Abstract

This article reflects on the role that technological change has played in the last century on international relations. It makes two main points. First, the relationship is reciprocal; while technological change has undeniable effects on international relations, the changing nature of world politics also affects the pace of technological change. Second, any technological change is also an exercise in economic redistribution and societal disruption. It creates new winners and losers, alters actor preferences, and allows the strategic construction of new norms and organizations. The nature of the technology itself, and the extent to which the public sector drives the innovation, generates differential effects on international relations. To demonstrate these arguments, special emphasis is placed on two important innovations of the last century for international relations: nuclear weapons and the Internet.

Keywords

innovation, international relations theory, Internet, liberalism, nuclear weapons, realism, technology

Technology's relationship with international relations in 2019 would appear to be transformative. Innovation has had salutary economic effects, increasing labor productivity and reducing extreme poverty across the globe. At the same time, new technologies have also created new threats. The destructive power of modern militaries would dwarf those of a century ago. Innovation also generates societal churn; anxiety about these changes has midwifed movements ranging from the Islamic State to #MeToo. Some technologies have led to radical shifts in the distribution of power among states, as countries like China and India have played catch-up to the advanced industrialized economies. The relationship between states and non-state actors has also been transformed, as anyone familiar with WikiLeaks or Facebook would acknowledge. A more

Corresponding author:
Daniel W Drezner, The Fletcher School of Law and Diplomacy, Tufts University, Medford, 160 Packard Avenue, MA 02155, USA
Email: daniel.drezner@tufts.edu

interconnected world has also highlighted vulnerabilities in the system, including exposure to pandemics such as Ebola. The speed of cross-border interactions has made diplomacy-by-tweet a thing. Despite a surge in protectionism from the G-20 economies, globalization continues apace. Technological change continues to lower the costs of cross-border exchange faster than governments can raise barriers.

As we mark the centenary of the academic study of International Relations in this issue, however, it is worth noting that almost every political effect of technology described in the previous paragraph would have applied with equal force in 1919. The First World War had just proven the awesome destructive capabilities of the machine gun and chemical weapons. A critical cause of that war was Germany's ability to catch up technologically to Great Britain, and German concerns with Russia's imminent catch-up. Non-state actors like the Suffragette movement, Second International, and Red Cross exploited new technologies to build transnational movements. Lowered transportation costs gave rise to the Spanish flu, the most deadly pandemic of the twentieth century. August 1914 was characterized by a flurry of rapid diplomatic and military signaling following the June assassination of Archduke Ferdinand. And in the run-up to the First World War, most European governments had raised tariffs in an effort to reduce economic interdependence. That did not stop trade from increasing; the spread of railroads and steamships overwhelmed protectionist barriers.[1] Perhaps the relationship between technological change and international relations might not be as transformative as technology enthusiasts believe.

Much IR scholarship treats technology as an exogenous shock, an independent variable that affects the contours of world politics through shifts in the distribution of military or economic power. There are two significant problems with this assumption, however. First, the causal arrow can also run in the other direction. Changes in the international system can also have pronounced effects on the pace of technological change. Second, not all kinds of technological change are created equal. Indeed, the development of nuclear weapons and the Internet has had wildly different effects on the the international system.

This article reflects on the role of technology has played in the last century of international relations. The economics of technological change are overwhelmingly positive, which can lull social scientists into a straightforward rationalist account of how great powers think about it. The international politics are far more fraught. Any technological change is also an exercise in redistribution. It can create new winners and losers, alter actor preferences, and allow the strategic construction of new norms. The nature of the technology itself, and the degree to which the public sector drives innovation, generates differential effects on international relations. To examine these dynamics, special emphasis is placed on two important innovations of the last century for international relations: nuclear weapons and the Internet.

Rationalist accounts of technological chance ... and their discontents

The salience of technology to world politics is often reduced to the simple equation that for sovereign actors, more technological innovation equals more power and plenty.

Economists stress the unique importance of technological change to economic growth. Most econometric studies find that technological innovation is responsible for anywhere between 60 and 85 percent of economic growth in the developed world.[2] The diffusion of technology to less advanced economies is equally important. According to one recent literature review, foreign sources of technology are responsible for approximately 90 percent of productivity growth in most countries.[3] Regardless of where countries lie on the distribution of power or economic development, their primary engine of economic growth comes through technology. And it is through economic growth that states can convert plenty into power.

Because of technology's manifest salience to maximizing wealth and security, many scholars adopt a rationalist take on how actors will promote innovation. This is most evident in the economics literature, which focuses on domestic economic variables to explain the variation in technological innovation. Factors such as the availability and sophistication of capital markets, size of the market, or allocation of entrepreneurial ability are considered to be the causal drivers. This is true even when economists consider the role of states. The economic historian David Landes, in explaining why China did not maintain its technological edge over Europe, argued 'China lacked a free market and institutionalized property rights' and that 'the Chinese state was always stepping in to interfere with private enterprise'.[4]

Politics plays a minor role in these narratives. In part, this is because economists often view non-economic factors as epiphenomenal in their causal stories. For example, Gregory Clark has argued that institutions are unimportant in determining the pace of technological innovation. Eventually, once a state hits upon the correct institutional formula for promoting innovation, that institutional form will be adopted by all governments interested in promoting economic growth.[5] Clark's assumption is that once the correct institutions are seized upon, other actors will effortlessly switch to them.

Neorealists offer a causal logic that echoes economists. To them, the anarchy of the international system forces all great powers to exploit technological change or suffer the risk of not surviving. Kenneth Waltz and his acolytes posit that the self-help nature of the international system requires states to adapt to new technologies or face threats to their survival as they lag behind other states in relative capabilities.[6] For neorealists, the equation is simple: anarchy forces the rapid diffusion of technology across the world. Emily Goldman notes that 'Most neorealists posit few barriers to diffusion, short of the need for adequate resources and information'.[7] Both neorealists and neoclassical economists agree that the incentives to acquire new technologies dominate all other factors.

This is not how most political scientists think about institutional change. There are several problems with frictionless theories of technological innovation and diffusion. The most obvious is that they ignore the ways that international relations is an independent variable affecting the pace of technological innovation. Consider, for example, Joseph Schumpeter's five canonical categories of innovation: invention, innovation in production processes, finding new markets, discovering new sources of supply, and developing new modes of economic organization.[8] They are only partially about technology itself; they are also about how societies respond to new opportunities presented by technological change. Three of Schumpeter's category of innovations are mediated through international relations to some degree.

Some economic historians acknowledge the role of international relations as an independent variable affecting the pace of innovation. They reference the distribution of power to explain why the Industrial Revolution originated in Europe rather than China. Both economic and military historians cite the anarchical politics of post-Westphalian Europe as a key permissive condition, in contrast to China's regional hegemony in East Asia. Joel Mokyr noted:

> the West was politically fragmented between more or less autonomous units that competed for survival, wealth, and power … The struggle for survival guaranteed that in the long run rulers could not afford to be hostile to changes that increased the economic power of the realm because of the real danger that an innovation or innovator would emigrate to benefit a rival.

Paul Kennedy similarly observed, 'The warlike rivalries among [Western Europe's] various kingdoms and city-states stimulated a constant search for military improvements'.[9]

The inference to draw is that there is a 'sweetspot' among the possible distributions of power that incentivizes states to invest in technological innovation. According to this logic, unipolarity depresses the rate of technological innovation. The hegemon feels so secure that it has little incentive to invest in disruptive innovations. Other states see the gap between themselves and the hegemon as too great to be overcome, blunting efforts to innovate. Similarly, a distribution consisting only of small states facing the threat of extinction would be too focused on the short term to bother with technological investments that pay out in the far future. Multipolarity, on the other hand, would generate the right mix of security and insecurity to spur more rapid innovations.

This is the logic that multiple scholars have used to explain why the Industrial Revolution originated in Great Britain.[10] Because it was an island, the country possessed a greater degree of security than the continental European states. At the same time, Great Britain was a part of the European state system. To maintain its relative position, it had an incentive to invest in science and technology as well.

Other IR approaches privilege the domestic politics of leaders and laggards as the key explanatory variable. Both long cycle and power-transition theorists stress the domestic conditions of an aspiring hegemonic power as determining their relative rise. Countries acquire hegemonic status because they create fertile conditions to develop a cluster of technologies in leading sectors. Acemoglu and Robinson, for example, argue that innovation is more likely to occur in governments with inclusive selectorates. More authoritarian governments are warier of innovations that could upset the domestic status quo. This is why, for example, Tsarist Russia proved to be a reluctant adopter during the Industrial Revolution. More open polities, such as Great Britain and the United States, emerged as technological hegemons.[11]

Radical innovations in one sector generate spillover effects for the entire lead economy. This, in turn, leads to diffusion of growth and technological change across the global economy. Over time, however, these 'technological hegemons' fail to maintain the rate of innovations, a phenomenon that Joel Mokyr labeled Cardwell's Law: 'no nation has been (technologically) very creative for more than a historically short period'.[12]

The reasons proffered for stagnation within the hegemonic actor are often political in nature. The most prominent political economy argument is that innovation represents a threat to those with a vested interest in the status quo. Schumpeter's aphorism about 'creative destruction' is commonly understood today as meaning that the value-added of new technologies dramatically outweighs the costs. This does not change the fact that innovation can also lead to massive redistributions of wealth. Those who stand to lose have an incentive to lobby the state to prevent relative losses. For Mancur Olson, the paradox of political stability leads to an accrual of rent-seeking that blunts technological dynamism. For Robert Gilpin, the domestic politics of a complacent hegemon lent itself to promote consumption rather than investment in the necessary public goods for innovation. For Paul Kennedy, hegemons suffered the wages of 'imperial overstretch', diverting scarce capital from needed investments in technology. For Daron Acemoglu, the technological leader has to play the thankless role of 'cutthroat capitalist' to promote innovation, which in turn subsidizes the more generous social safety nets of supporter states.[13] Eventually, the cost of this public goods provision ceases to be political popular in the hegemonic state.

These explanations are not mutually exclusive, but they lead to the same outcome: a period of technological catch-up for challengers to the hegemon. Indeed, according to some economists, technologically backward countries have an advantage in catching up, as they can exploit their relative backwardness to leapfrog over technological dead-ends.[14] This period of stagnation and catch-up also triggers a period of geopolitical strife until a new hegemon emerges.[15]

The long cycle and diffusion literatures have different causal mechanisms than the decentralization argument. Mark Z. Taylor is likely correct when he argues that a synthesis of the two is most apt: 'countries for which external threats are relatively greater than domestic tensions should have higher national innovation rates than countries for which domestic tensions outweigh external threats'.[16]

These explanations for how international politics can drive technological change are essentially rationalist in nature. Other research, however, suggests that the role of prestige has been neglected as an explanatory factor in driving large-scale investments in science and technology.[17] Prestige is a positional good; one only benefits from it by outperforming other actors in a designated social setting. Historically, many of the social settings designated as the most important have been in the field of science and technology. This can lead actors to direct investments in technology projects that might have limited military or economic value but generate great prestige.

The most obvious example from the past century of prestige investments in technology would be the massive sums allocated for manned space programs by the Cold War superpowers. The USSR's 1957 launch of Sputnik and 1961 manned orbit of Earth had a pronounced effect on the United States and its prestige. These acts caused policymakers to worry that the United States was falling behind the Soviet Union in science and technology. Equally important, policymakers were concerned about the *perception* that the United States was falling behind. In October 1961, the US Information Agency commissioned surveys of European audiences on this question, and they found a strong majority of Europeans believed the USSR to be winning the space race. Washington expended considerable resources to catch up and overtake the Soviet Union. According to one estimate the US Apollo program accounted for 2.2 percent of all federal government spending at its peak, more than twice the level of the Manhattan Project.[18]

These kind of prestige-based investments in science and technology continue to affect twenty first–century investments in research and development. China's first manned space flight in 2003 triggered frenzied reactions from peer competitors, including a pledge by the United States to return to the moon and send a manned mission to Mars. Other countries with significant budget constraints nonetheless signaled interest in investing in outer space research. China's landing of a probe on the dark side of the moon in January 2019 generated similar responses.

The rationalist story of why states invest in technological innovation makes intuitive sense. But such a narrative is at best incomplete and at worst an incorrect description of how international politics affects technological change. The distribution of power, the nature of domestic institutions in leading economies, and the allure of prestige goods all drive innovation in ways different from the strictly rationalist account.

Varieties of technological change in world politics

Talking about technological change in the abstract runs the risk of treating all innovations similarly. Whether one uses an economic or military lens of evaluation, not all technologies are created equal. Even though military necessity was a driver of both inventions, for example, nuclear weapons have had very different effects on world politics than the Internet.

There are two criteria that help to parse how to think about the effect of different technologies on world politics. The first is the magnitude of fixed-cost investments necessary to develop or adopt a new innovation. Most inventions require significant fixed costs, but as a technology becomes more standardized, the investments necessary to develop them can decline.

That said, some technologies, even after standardization, still require considerable investments. This may be because the economic costs of production are high. In other cases, the disruptive effects of an innovation on political organizations or societal institutions are significant enough to make the costs of adaptation equally high. For example, the French navy pioneered several key technologies in the latter half of the nineteenth century, including shell guns and steel-hulled warships. Because it was unable to incorporate the organizational implications of those innovations, however, France found itself overtaken by other navies in the adoption of these technologies.[19]

The larger the fixed costs for acquiring innovations, the greater the barriers to entry for any actor attempting to acquire a new technology. The lower the fixed-cost investments, the greater number of expected actors who will be able to develop and consume the technology in question.

The other dimension is whether the primary innovator of a new technology caters to the public sector or the private sector. Most significant innovations qualify as 'general purpose' technologies that both the security and civilian sectors can exploit. Some technological innovations, however, have limited civilian applications. Nuclear fission, for example, proved extremely useful for military technology. As a source of civilian energy, however, its profitability is circumscribed, particularly during periods of low hydrocarbon prices. The principal sources of innovation in nuclear technology have come from the state sector. The Internet is a different story. While originally developed due to ample amounts of

	Public sector dominance	*Private sector dominance*
High fixed costs	**Prestige tech**	**Strategic tech**
Low fixed costs	**Public tech**	**General purpose tech**

Figure 1. A Typology of Technological Innovations.

government funding, the data flows from its commercial applications have far outstripped its official purposes. Most online innovation has come from the private sector.

This creates a simple 2 x 2 of the different kinds of technological innovations (see Figure 1).[20] Those inventions that have high fixed costs but significant civilian applications are *strategic tech*. Technologies with large fixed costs possess increasing-returns dynamics. This invites considerable amounts of great power intervention to promote national champions and avoid dependence on foreign technologies, akin to strategic trade theory. The civilian aircraft sector fits nicely within this quadrant, as does the current development of 5G networks.

Technologies that have large fixed costs and limited civilian applications fall under the *prestige tech* category. Only states have the incentive to develop these technologies. Some of these technologies, such as nuclear weapons, provide obvious utility to governments. All of the technologies in this box, however, function as prestige goods because of their cost. Supersonic transport planes like the Concorde would also fall squarely into this category, as would manned space exploration.

Technologies with low fixed costs but limited private-sector opportunities would be considered *public tech*. This label is apt because if the barriers to entry are low but private-sector interest is minimal, then the innovation is likely to possess strong public goods qualities. The private sector has less of an incentive to develop technologies that are both nonrival and nonexcludable.[21] If the technology offers considerable social benefits, however, then one could envision governments having an incentive to make the necessary investments. Public health innovations, like vaccines, would fall under this category.

Finally, those innovations with lower fixed costs and significant private-sector possibilities fall under the category of *general purpose tech*. These goods may have been developed with public-sector purposes in mind, but the commercial applications are so manifest that private-sector activity drives the direction of the technology. The most obvious current example of this kind of technology is the diffusion of drones – although the development of artificial intelligence could eventually fall under this category as well.[22]

This typology is useful in considering how quickly certain technologies will diffuse from invention to standardization, and how great powers are likely to react to new innovations that can affect world politics. The more that a technology approaches the general purpose category, the more quickly it will diffuse across the globe. The lower the fixed costs – whether material, organizational, or societal in nature – the more rapidly a technology should diffuse from leader to laggards. Therefore, general purpose tech should diffuse the quickest, whereas prestige tech should have the most limited degree of proliferation.

Public policy also plays a critical role in determining the degree of diffusion. States have multiple domestic and international incentives to regulate the diffusion of new innovations. Governments on the technological frontier will be more likely to advocate for strong intellectual property rights. In theory, this should incentivize further innovations. The knock-on effect of such a policy is to limit the diffusion of technology across borders.[23] Similarly, governments are also likely to restrict access to technologies that have significant effects on military capabilities. This applies to nuclear weapons; there is an entire regime complex devoted to limiting the proliferation of that technology.

How technology affects international relations

The rest of this article will focus on two prominent technologies of the past century: nuclear weapons and the Internet. This is not because they are typical, but rather because they represent most-different forms of technological change. Nuclear weapons qualify as prestige tech, while Internet applications primarily fall within general purpose tech. Exploring the effects of these two significant innovations on different dimensions of international relations theory should offer some fruitful reflections.

There are a limited number of components to international relations theory. Different theories stress different elements, but paradigms ranging from realism to poststructuralism have to cope with concepts like power, interest, and norms. Technological change has affected each of these dimensions of world politics. This section considers how the innovation of nuclear weapons and the Internet have altered perceptions of each of these concepts.

Power

Nuclear weapons have existed since 1945, but as a form of prestige tech their diffusion has been restricted to fewer than ten actors. Despite this limited diffusion, international relations theorists have concluded that their development has had system-transforming effects. Thomas Schelling noted in the mid-1960s, ‘our era is epitomized by words like “the first time in human history” and by the abdication of what was “permanent”’.[24] Once all the great powers became nuclear-armed nations, a large strain of international relations scholarship was dedicated to thinking about how nuclear weapons altered the dynamics of world politics.

This is most pronounced within the realist paradigm. Kenneth Waltz’s structural realism was predicated on the notion that the structure of anarchy has been a constant in global affairs for centuries. Nonetheless, as the Cold War was ending, he concluded in multiple articles that nuclear weapons had altered the nature of the international system:

> Never since the Treaty of Westphalia in 1648, which conventionally marks the beginning of modern history, have great powers enjoyed a longer period of peace than we have known since the Second World War. One can scarcely believe that the presence of nuclear weapons does not greatly help to explain this happy condition.[25]

How did nuclear weapons affect international politics so profoundly? This innovation transformed power politics by disconnecting the power to destroy with other capabilities previously thought to be necessary for great power status. Once a country possesses a perceived second-strike capability – either through hardened defenses or uncertainty about the location of their nuclear stockpiles – they possess the power to deter. There is no correlation, however, between nuclear deterrence and conventional measures of military power. As Waltz notes, 'Nuclear weapons purify deterrent strategies by removing elements of defense and warfighting'. Schelling is even more stark on this point: 'Nuclear weapons make it possible to do monstrous violence to the enemy without first achieving victory'.[26] At the same time, because nuclear weapons accomplish nothing but destruction, their utility in other arenas of world politics is limited. The power of nuclear weapons is not terribly fungible, even when compared with other dimensions of military power.

A paradoxical effect of the development of nuclear weapons is that it disconnects measurements of state power from other areas of technological innovation. Although the barriers to entry in the nuclear club are high, technologically backward states have been able to surmount it. Maoist China became a nuclear-capable state when it was one of the poorest countries on Earth. Neither a fractious Pakistan nor an autarkic North Korea were thought to be on the technological frontier, and yet these countries were able to develop nuclear warheads and the ballistic missile technology necessary to deliver these weapons. Once a state possesses nuclear weapons; however, they are afforded a different place in the power hierarchy than otherwise would be the case. This is true even in instances in which a state lags behind in other areas. As Waltz notes, 'So long as a country can retaliate after being struck, or appears to be able to do so, its nuclear forces cannot be made obsolete by an adversary's technological advances'.[27] This explains how Russia is viewed as one of the great powers in the twenty-first century despite controlling less than 4 percent of the global economy. This is why Iran and North Korea have occupied so much great power attention for the last few decades.

While nuclear warheads are prestige tech, the Internet and its myriad applications are general purpose tech. The commercialization of cyberspace generated two initial strands of scholarly thought about its effect on power. The first was that the Internet altered the balance of power between states and non-state actors. In terms of communication speed, the Internet was not much faster than the telegraph. What made the Internet different was its low cost and resultant ease of access. Cyberspace's low barriers to entry dramatically lowered the transaction costs of non-state actors to organize. Furthermore, the Internet's creators viewed state efforts to disrupt or prevent online activism with profound skepticism. The decentralized architecture of the Internet ostensibly made it impossible for states to regulate. John Perry Barlow famously told *Time* magazine in 1993, 'The Net interprets censorship as damage and routes around it'. The use of social media and other online applications in color revolutions and the Arab Spring lent credence to this argument.

The second strand of thought was that the Internet empowered liberal democracies over other regime types in world politics. Cyberspace originated with the United States and appeared to prize openness. It was viewed as enhancing all facets of America's soft power. The explosion of content in cyberspace demonstrated the vibrancy of American culture, as well as the US commitment to liberal values of open discourse. Its development was also a credit to prior US policies and would also augment American soft power.[28] Cyberspace's norm of openness also made it logical that democracies would benefit the most. Helen Milner argued that the nature of a country's democratic institutions determined how quickly it adopted online technologies. Writing in 2006, she concluded that democracies were far more likely to facilitate the spread of online tech than authoritarian regimes.[29]

Recent scholarship and recent events offer a more muddled picture. States learned how to be able to regulate the Internet to serve their national interest with relative rapidity. On issues ranging from content regulation to copyright to privacy, great power governments were able to get their way in cyberspace.[30] Domestically, authoritarian regimes moved quickly down the leaning curve in permitting Internet access while still being able to regulate the flow of information.[31] Initial scholarship underestimated the ways in which top-down bureaucracies could harness online data to exercise greater control over civil society. These actions have not eliminated the ability of non-state actors to influence world politics. But greater state capacity has likely redirected where non-state actors have had their greatest influence. WikiLeaks, for example, has focused its energies on Western democracies far more than authoritarian great powers such as Russia or China.

Furthermore, in recent years, authoritarian states have become more adept at exploiting online technologies to attack liberal democracies. North Korea responded to the release of a film satirizing Kim Jong Un by hacking into Sony Pictures and publishing embarrassing emails from its executives. The fallout cost Sony tens of millions of dollars. China exploited its cyber-capabilities to steal billions in intellectual property from the Western democracies. Russia's use of troll farms, sophisticated hacking, and disinformation campaigns to influence elections in the West is well-known. Over time, revisionist actors have become adept at using these techniques to counteract the soft power of the West.

The prestige tech of nuclear weapons augmented the power of those actors who developed them, widening power inequalities between nuclear-armed states and non-nuclear states. The general purpose tech of cyberspace initially benefited liberal democratic actors more than others. Over time, however, its diffusion and exploitation by revisionist actors had a leveling effect on the balance of power.

Interests

An ongoing scholarly debate is whether actors view the international system as a zero-sum or nonzero-sum game. This difference in preferences is at the root of differences between realist and liberal theories of international politics. How states reacted to the development of nuclear and online technologies offer a useful guide into how technologies can alter or reveal changes in state preferences.

What is striking about the effect of nuclear weapons on world politics is that it falsified the notion that world politics was a zero-sum enterprise. Scholars of the nuclear age agree that the United States held an overwhelming superiority in nuclear weapons in their first decade of existence. The United States had demonstrated no compunction about using atomic weapons to hasten the end of the Second World War. If, as realists claim, world politics is a strictly zero-sum game, then the United States should have used its weapons in the 1950s to launch a preemptive strike against the Soviet Union. Given its advantages in the arms race at that time, Washington would have secured itself a relative advantage in any ensuing conflict. At a minimum, it should have been willing to use its nuclear stockpile to secure victory in the Korean War.

Of course, the use of nuclear weapons in the 1950s is a non-event. While the United States had internal debates about deploying atomic weapons in the Korean War, in the end policymakers ruled out any first use.[32] The question of a preemptive strike on the Soviet Union was barely discussed. President Eisenhower recognized that nuclear preemption was a viable option. In his diary, however, he rejected the notion as contravening US traditions and procedures.[33]

It is also striking to see how the emergence of nuclear weapons triggered a surprisingly strong regime complex devoted to arms control and nonproliferation. Beginning with the 1963 Limited Nuclear Test Ban Treaty, the United States and Russia agreed to a welter of arms control agreements limiting their nuclear stockpiles by warhead yield, type of delivery vehicle, and related systems. The rest of the world agreed to the Nuclear Non-Proliferation Treaty (NPT) in 1970, and then permanently extended it in 1995. The International Atomic Energy Agency was created to enforce the NPT. In recent decades, an array of supporting and reinforcing agreements, including the Proliferation Security Initiative and the Nuclear Safety Summits, has also emerged. This kind of regime complex does not resemble the zero-sum description that is often ascribed to security issues. It would appear that the destructive power of nuclear weapons triggered higher levels of international cooperation than would otherwise have been expected. Indeed, the regime complex for nuclear weapons is far more robust than the one for conventional weaponry.

As general purpose tech, the Internet was expected to jumpstart increases in labor productivity and economic growth. It would therefore be unsurprising to expect a more cooperative, nonzero-sum approach to its regulation. Intriguingly, this has not been the case. A quick survey of negotiations over different aspects of Internet regulation shows that national preferences and bargaining cleavages mirrored offline policy disputes.[34] The United States and European Union have consistently been at loggerheads over questions of competition policy and privacy rights. As China and Russia have embraced the authoritarian uses of online activity, debates over Internet governance have become far more contentious.

There are several possible explanations for this failure of online innovations to alter national preferences significantly. Since cyberspace's effect is as much on state-society relations as interstate relations, national governments prioritize domestic political considerations in the articulation of policy preferences. While all states are interested in the commercial opportunities afforded by online commerce, the past two decades demonstrate that national and internal security considerations will have priority. It is therefore unsurprising that issues like content regulation have no international agreement.

Another possible explanation is that the Internet's anticipated productivity boom has not come to pass. The one economic trend associated with the development of online technologies has been the emergence of natural monopolies. Firms like Facebook, Google, and Weibo profit from considerable network externalities, achieving dominance in their particular fields. The cumulative macroeconomic effect of this kind of corporate concentration is to drag down economic growth.[35] To date, the actual effects of the Internet on economic growth in the developed world have been far less than advertised.[36]

The effect of both prestige tech and general purpose tech on actor interests in world politics is indeterminate. The destructive power of nuclear weapons led to more cooperation than a zero-sum set of preferences would have predicted. Cooperation on nuclear weapons is a data point for liberals, demonstrating that complex interdependence can lead to cooperation in unlikely areas.[37] At the same time, the positive-sum nature of online innovation has not necessarily led to more cooperation in cyberspace. Conflict in cyberspace is a data point for realists, suggesting that greater interdependence breeds greater conflict.[38]

Norms

Technology has been hypothesized to have a direct effect on the spread of norms in the international system. In discussing the phenomenon of norm cascades, Martha Finnemore and Kathryn Sikkink explicitly tied the pace and scope of norm acceptance to technological change:

> Changes in communication and transportation technologies and increasing global interdependence have led to increased connectedness and, in a way, are leading to the homogenization of global norms … the speed of normative change has accelerated substantially in the later part of the twentieth century.[39]

This hypothesis would seem to hold with even greater force in the twenty-first century. Norms regarding violence against women and gay marriage, for example, spread more quickly than similar twentieth century norms.

Technological innovation has certainly facilitated the spread of norms. A more complex question is the role that norms play in the promulgation and use of new technologies. Almost by definition, new technologies emerge in environments that lack indigenous norms. The effect of a new technology on the social world is inherently uncertain. This uncertainty can make it difficult to envision which norms, rules, and codes of conduct are appropriate. New innovations might try to copy norms and practices from adjacent technologies, but whether those norms will apply with equal force to a new technological arena is an open question. Attempting to analogize old norms to new areas can lead to significant misperceptions.[40] As Finnemore and Sikkink noted, the promulgation of new norms is a strategic act.

This dynamic has been on display in the evolution of norms regarding of nuclear weapons and cyberspace. Initially, the invention of the atomic bomb did not alter US military doctrine. Indeed, their use in August 1945 demonstrates that US decision-makers did not view these weapons differently from other forms of ordinance. As Nina Tannenwald notes, 'the atomic bombings were simply an extension of strategic bombing during a war that had substantially elevated the scale of destruction, flattening great cities in Europe and Japan'.[41]

As the destructive power of these weapons became manifestly clear, however, a 'nuclear taboo' emerged against the first use of these weapons.[42] This was materially harmful to the United States. In the 1950s, there were internal policy debates about using nuclear forces in the Korean War. More generally, the Soviet Union's conventional military superiority in Central Europe necessitated a US doctrine of first-use in case of a Warsaw Pact attack on North Atlantic Treaty Organization (NATO) forces.

The Eisenhower administration engaged in vigorous rhetorical efforts to make nuclear threats normatively acceptable in world politics. US officials found themselves on the losing side of the argument, even though there is suggestive evidence that nuclear threats are effective in international relations.[43] Beginning with the United Nations construction of the 'weapons of mass destruction' category, this technology was perceived as being radically different from other forms of ordinance. This was true even though conventional strategic bombing had incurred a much greater loss of life in the Second World War. A welter of non-state actors pushed the taboo rhetoric, aided by a Soviet Union that at the time was disadvantaged in the nuclear arms race. Continued innovations in nuclear weapons technology only reinforced the taboo. As their theoretical destructive power increased, the use of these weapons became less and less conceivable. By the end of Eisenhower's second term, he acknowledged that 'The new thermonuclear weapons are tremendously powerful; however, they are not … as powerful as is world opinion today in obliging the United States to follow certain lines of policy'.[44]

By the end of the Cold War, the taboo had been internalized by all the great powers. As Tannenwald notes, 'The taboo was no longer simply a "constraint" but had itself become a foreign policy goal for a "civilized" state'.[45] This can be seen in debates about the use of low-yield tactical nuclear weapons. Even though these weapons could have been viewed as being useful during armed conflicts like the gulf wars, their actual use was eventually ruled out.[46]

Norms are not set in stone, and recent research has focused on the myriad tactics and strategies that can be deployed to alter or constrain a particular norm.[47] This parallels efforts by the two largest nuclear states to gain some degree of tactical flexibility within the nuclear taboo. This century has seen the end of the ABM and INF treaties, weakening the overall arms control regime. Russia has recently altered its nuclear doctrine to lower the threshold for the use of low-yield weapons. In response, the United States has also changed its doctrine to 'enhance the flexibility and range of its tailored deterrence options … for the preservation of credible deterrence against regional aggression'.[48] Whether this shift in doctrine leads to a shift in norms is unclear. Nonetheless, nuclear weapons are unique as a technology that defines great power relations even though they have not been used in more than 70 years.

As with nuclear weapons, the norms environment surrounding cyberspace has evolved over time. On governance issues, for example, the initial debates surrounding how to run the Internet echoed the norm of multistakeholderism, familiar to academic researchers that created the architecture of cyberspace.[49] By the late 1990s, what emerged was a multi-stakeholder model built on public–private partnerships. As Jeffrey Lantis and Daniel Bloomberg characterized it, 'the goal was to establish a decentralized network for cyberspace that would be "democratic" and treat the views of major corporations, governments, civil society, and even academics as equally important to help establish standards for the Internet activity'.[50]

While multistakeholderism explained the norms governing early Internet governance, the norms governing cybersecurity looked very different from the outset. Security concerns about cyberspace emerged even before the Internet became general purpose tech. As early as 1991, a US National Research Council study warned, ‘Tomorrow’s terrorist may be able to do more damage with a keyboard than with a bomb’.[51] From the early days of trying to theorize about cybersecurity, researchers and defense strategists imported wholesale the norms of deterrence from nuclear doctrine.[52] According to this logic, the prospect of catastrophic cyberattacks required states to communicate the likelihood of retaliation.

Over time, however, initial governance and cybersecurity norms have come under severe challenge. As the Internet diffused, countries such as Russia and China began to lobby for governance norms to shift from multistakeholderism to multilateralism. To that end, they lobbied for control over Internet governance to be shifted from Internet Corporation for Assigned Names and Numbers (ICANN) to the International Telecommunications Union (ITU), a more conventional international governmental organization. This led to a series of contentious ITU conferences in Dubai and Busan in the 2010s. At these meetings, the United States and Western allies functioned as ‘antipreneurs’, creating powerful points of resistance to changing overall Internet governance.[53]

On cybersecurity, the deepening of online activity could have strengthened the deterrence norm. As the Internet diffused across the globe, the developments of cloud computing, social media, and smartphones meant that more aspects of daily life became intertwined with the online world. The greater degree of interconnectedness also meant that cyberattacks had the potential to pose a greater threat to economic growth and national security.

In recent years, however, it has become clear that the deterrence analogy has numerous flaws. The scope of cyberattacks has not been even remotely as significant as those involving nuclear weapons. Empirical research into actual cyberattacks reveals minimal negative feedback on bilateral interstate relationships.[54] Furthermore, the question of attribution dominates questions about cyberattacks. Cyberspace is a complex entity involving an array of state actors and networked non-state actors. The success of groups like WikiLeaks, Anonymous, and LulzSec, with varying degrees of great power ties, confirms that this security environment resembles espionage far more than nuclear security. As Tim Stevens notes, ‘the conditions pertaining in cyberspace are such that it has been difficult to transfer the procedures and techniques of Cold War deterrence to this domain’.[55]

The United States, in response to the failures of deterrence, has tried a different normative approach. In 2010, the US Deputy Secretary of Defense noted that ‘If there are to be international norms of behavior in cyberspace, they may have to follow a different model [than nuclear arms control], such as that of public health or law enforcement’.[56] Joseph Nye argues that crime is the superior analogy to Cold War images of deterrence.[57] NATO’s development of the Tallinn Manual, as well as the US use of legal norms to threaten prosecution of Chinese hackers, appears to be yielding more success at the promulgation of norms than the nuclear deterrence norms.

Finally, the proliferation of social media has also altered diplomatic norms. The progenitors of the modern Internet have been aware of this possibility for decades. In 1985, researchers warned that online discourse was different from other forms of communication because of its 'propensity to evoke emotion in the recipient … and the likelihood that the recipient will then fire off a response that exacerbates the situation'. The analysts concluded 'These media are quite different from any other means of communication. Many of the old rules do not apply'.[58] This accurately captures the early twenty-first century, in which great power rivals troll each other on Twitter.[59]

The evolution of norms within technologically innovative areas has varied in different area of tech. With nuclear weapons, the emergence of a strong taboo against their use suggests a standard constructivist model of norms being deepened and internalized. With cyberspace, however, the initial set of norms imported from other arenas has not worked terribly well. As a result, cyber norms continue to be in flux.

Conclusion

This article has considered how technological innovation has affected international relations and vice versa. Contrary to strictly rationalist accounts, culture and prestige also affect patterns of innovation and diffusion. Technology is not merely an independent variable in international relations theory. There are reciprocal effects.

Regardless of the kind of tech, significant innovations augment the power of the actors that harness those inventions. Whether that power advantage persists depends on the quality of the technology itself. General purpose tech has a greater leveling effect than prestige tech. Technology's effects on national interests are contradictory. The prestige tech of nuclear weapons engendered surprising levels of cooperation, while cyberspace has triggered surprising levels of conflict. Prestige tech created norms that only grew more powerful over time. General purpose tech, on the other hand, developed a far more contentious normative environment. The pace of innovation within general purpose tech might be so accelerated that norm creation and norm maintenance becomes more difficult.

Concerns about technology's ability to create new threats are hardly a new phenomenon. On the other hand, technologies that look uncontrollable when they first emerge, look sedate with time. In 2019, the notion of diplomacy-by-tweet seems strange; a decade from now, it might seem as quaint as diplomatic cables. It might be more accurate to say that innovation can create temporary uncertainties in world politics. Over time, human beings adapt.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

Notes

1. Kevin O'Rourke and Jeffrey Williamson, *Globalization and History* (Cambridge, MA: The MIT Press, 1999).

2. Elhanan Helpman, *The Mystery of Economic Growth* (Cambridge, MA: Harvard University Press, 2004).
3. Wolfgang Keller, 'International Technology Diffusion', *Journal of Economic Literature*, 42(3), 2004, pp. 752–82.
4. David Landes, 'Why Europe and the West? Why Not China?' *Journal of Economic Perspectives*, 20(3), 2006, pp. 3–22.
5. Gregory Clark, *A Farewell to Alms: A Brief Economic History of the World* (Princeton, NJ: Princeton University Press, 2007).
6. Kenneth Waltz, *Theory of International Politics* (Reading, MA: Addison-Wesley, 1979).
7. Emily Goldman, 'Cultural Foundations of Military Diffusion', *Review of International Studies*, 32(1), 2006, pp. 69–91.
8. Joseph Schumpeter, *Capitalism, Socialism, and Democracy* (New York: Harper & Brothers, 1942).
9. Joel Mokyr, *The Lever of Riches: Technological Creativity and Economic Progress* (New York: Oxford University Press, 1990), p. 206; Paul Kennedy, *The Rise and Fall of the Great Powers* (New York: Vintage Press, 1987), p. 2.
10. Mokyr, *Lever of Riches*.
11. Daron Acemoglu and James Robinson, *Why Nations Fail: The Origins of Power, Prosperity, and Poverty* (New York: Broadway Business, 2013).
12. Mokyr, *Lever of Riches*, p. 207.
13. Mancur Olson, *The Rise and Decline of Nations* (New Haven, CT: Yale University Press, 1982); Robert Gilpin, *War and Change in World Politics* (New York: Cambridge University Press, 1981); Kennedy, *Rise and Fall of the Great Powers*; Daron Acemoglu, James A. Robinson, and Thierry Verdier, 'Asymmetric Growth and Institutions in an Interdependent World', *Journal of Political Economy*, 125(5), 2017, pp. 1245–305.
14. Alexander Gerschenkron, *Economic Backwardness in Historical Perspective* (Cambridge, MA: Belknap Press, 1962).
15. Daniel W. Drezner, 'State Structure, Technological Leadership and the Maintenance of Hegemony', *Review of International Studies*, 27(1), 2001, pp. 3–25.
16. Mark Z. Taylor, 'Toward an International Relations Theory of National Innovation Rates', *Security Studies*, 21(1), 2012, pp. 113–52.
17. Lilach Gilady, *The Price of Prestige: Conspicuous Consumption in International Relations* (Chicago, IL: University of Chicago Press, 2018).
18. Paul Musgrave and Daniel Nexon, 'Defending Hierarchy from the Moon to the Indian Ocean: Symbolic Capital and Political Dominance in Early Modern China and the Cold War', *International Organization*, 72(3), 2018, pp. 591–626.
19. Michael Horowitz, *The Diffusion of Military Power* (Princeton, NJ: Princeton University Press, 2010).
20. See also Horowitz, *The Diffusion of Military Power*, p. 33.
21. Olson, *Rise and Decline of Nations*. One possibility is if firms bundle innovations with public goods qualities with products that do confer excludable benefits to them. Both web browsers and social media sites would fall under this category, in which network externality effects are far more powerful than the free-riding effects from traditional public goods.
22. On drones, see Sarah Kreps, *Drones: What Everyone Needs to Know* (New York: Oxford University Press, 2016); on AI, see Michael Horowitz, 'Artificial Intelligence, International Competition, and the Balance of Power', *Texas National Security Review*, 1(3), 2018, pp. 37–57.
23. Susan Sell, *Power and Ideas* (Albany, NY: SUNY Press, 1998).
24. Thomas Schelling, *Arms and Influence* (New Haven, CT: Yale University Press, 1966), p. 18.

25. Kenneth Waltz, 'Nuclear Myths and Political Realities', *American Political Science Review*, 84(3), 1990, pp. 731–45.
26. Waltz, 'Nuclear Myths and Political Realities', p. 732; Schelling, *Arms and Influence*, p. 18.
27. Kenneth Waltz, 'The Emerging Structure of International Politics', *International Security*, 18(2), 1993, pp. 44–79.
28. Joseph Nye, *The Future of Power* (New York: PublicAffairs, 2011).
29. Helen V. Milner, 'The Digital Divide: The Role of Political Institutions in Technology Diffusion', *Comparative Political Studies*, 39(2), 2006, pp. 176–99.
30. Daniel W. Drezner, 'The Global Governance of the Internet: Bringing the State Back in', *Political Science Quarterly*, 119(3), 2004, pp. 477–98.
31. Bruce Bueno De Mesquita and George W. Downs, 'Development and Democracy', *Foreign Affairs*, 84(5), 2005, pp. 77–86.
32. Nina Tannenwald, 'The Nuclear Taboo: The United States and the Normative Basis of Nuclear Non-Use', *International Security*, 53(3), 1999, pp. 433–68.
33. David Alan Rosenberg and W. B. Moore, 'Smoking Radiating Ruin at the End of Two Hours': Documents on American Plans for Nuclear War with the Soviet Union, 1954–55', *International Security*, 6(3), 1982 [1981], pp. 3–38.
34. Drezner, 'Global Governance of the Internet'.
35. Jan De Loecker and Jan Eeckhout, 'The Rise of Market Power and the Macroeconomic Implications' (NBER working paper 23687, August 2017), available at: http://www.janeeckhout.com/wp-content/uploads/RMP.pdf
36. Robert Gordon, *The Rise and Fall of American Growth* (Princeton, NJ: Princeton University Press, 2016).
37. Robert Keohane and Joseph Nye, *Power and Interdependence* (Boston, MA: Longman, 1977).
38. Waltz, *Theory of International Politics*.
39. Martha Finnemore and Kathryn Sikkink, 'International Norm Dynamics and Political Change', *International Organization*, 52(4), 1998, pp. 887–917.
40. Yuen Foong Khong, *Analogies at War* (Princeton, NJ: Princeton University Press, 1992).
41. Tannenwald, 'The Nuclear Taboo', p. 442.
42. Much of this paragraph draws from Nina Tannenwald, 'Stigmatizing the Bomb: Origins of the Nuclear Taboo', *International Security*, 29(4), 2005, pp. 5–49.
43. Robert Pape, *Bombing to Win* (Ithaca, NY: Cornell University Press, 1996).
44. Tannenwald, 'Stigmatizing the Bomb', p. 27.
45. Tannenwald, 'The Nuclear Taboo', p. 460.
46. Tannenwald, 'The Nuclear Taboo', pp. 458–62.
47. Rebecca Sanders, 'Norm Spoiling: Undermining the International Women's Rights Agenda', *International Affairs*, 94(2), 2018, pp. 271–91.
48. U.S. Department of Defense, '2018 Nuclear Posture Review', available at: https://media.defense.gov/2018/Feb/02/2001872886/-1/-1/1/2018-NUCLEAR-POSTURE-REVIEW-FINAL-REPORT.PDF (accessed 1 June 2018), p. xii. On Russia's changing nuclear doctrine, see Nikolai Sokov, 'The Origins of and Prospects for Russian Nuclear Doctrine', *Nonproliferation Review*, 14(2), 2007, pp. 207–26; Nina Tannenwald, 'How Strong is the Nuclear Taboo Today?' *The Washington Quarterly*, 41(3), 2018, pp. 89–109.
49. Mark Raymond and Laura DeNardis, 'Multistakeholderism: Anatomy of an Inchoate Global Institution', *International Theory*, 7(3), 2015, pp. 572–616.
50. Jeffrey Landis and Daniel Bloomberg, 'Changing the Code? Norm Contestation and US Antipreneurism in Cyberspace', *International Relations*, 32(2), 2018, pp. 149–72.

51. Quoted in Eric Sterner, ‘Retaliatory Deterrence in Cyberspace’, *Strategic Studies Quarterly*, 5(1), 2011, pp. 62–80.
52. Tim Stevens, ‘A Cyberwar of Ideas? Deterrence and Norms in Cyberspace’, *Contemporary Security Policy, 33*(1), 2012, pp. 148–70; Joseph Nye, ‘Deterrence and Dissuasion in Cyberspace’, *International Security*, 41(3), 2017 [2016], pp. 44–71.
53. Lantis and Bloomberg, ‘Changing the Code?’
54. See, for example, Brandon Valeriano and Ryan C Maness, ‘The Dynamics of Cyber Conflict Between Rival Antagonists, 2001–11’, *Journal of Peace Research*, 51(3), 2014, pp. 347–60; Maness and Valeriano, ‘The Impact of Cyber Conflict on International Interactions’, *Armed Forces & Society*, 42(2), 2016, pp. 301–23; Erik Gartzke, ‘The Myth of Cyberwar’, *International Security*, 38(2), 2013, pp. 41–73.
55. Stevens, ‘A Cyberwar of Ideas?’ p. 151.
56. William J. Lynn III, ‘Defending a New Domain: The Pentagon’s Cyberstrategy’, *Foreign Affairs*, 89(5), 2010, pp. 97–108.
57. Nye, ‘Deterrence and Dissuasion in Cyberspace’, p. 45.
58. Norman Shapiro and Robert Anderson, ‘Toward an Ethics and Etiquette for Electronic Mail’ (RAND report R-3283-NSF/RC, July 1985), available at: http://www.rand.org/pubs/reports/R3283/index1.html (accessed 15 June 2018).
59. Daniel W. Drezner, ‘How Trolling Could Become the New International Language of Diplomacy’, *Washington Post*, 15 May 2015, available at: https://www.washingtonpost.com/opinions/how-trolling-could-become-the-new-international-language-of-diplomacy/2015/05/15/5b092014-f9a0-11e4-a13c-193b1241d51a_story.html?noredirect=on&utm_term=.b1d545df30af

Author biography

Daniel W Drezner is Professor of International Politics at the Fletcher School of Law and Diplomacy at Tufts University. His recent work examines the interplay between power and ideas in international politics. Drezner’s publications include *All Politics is Global* (Princeton, 2007) and *The Ideas Industry* (Oxford, 2017).